MolPallete — Corpus EDA

naveja_recap

/home/mogan/preprocessed/molpallete/coconut-only/naveja_recap

generated 2026-08-31 12:29

A · Provenance

field	value
sources	['coconut']
method	naveja_recap
decomposition_params	{'ratio': 0.3333333333333333, 'include_ring': True, 'max_cores':
core_ratio	0.3333333333333333
max_cores	4
max_rgroups	8
keep_stereo	True
neutralise	False
dedup	True
n_duplicates_removed	330
size_filter	{'min_heavy_atoms': 5, 'max_heavy_atoms': 50}
graph_hash_version	4
rdkit_version	2026.03.2
created_at	2026-08-31T09:27:59

B · Composition

quantity	value
molecules	374,454
from coconut	374,454 (100.0%)
decompositions	1,085,929
per molecule	mean 2.90, median 3, max 4
R-group slots	1,536,136
per decomposition	mean 1.41
k >= 2	31.79% of decompositions

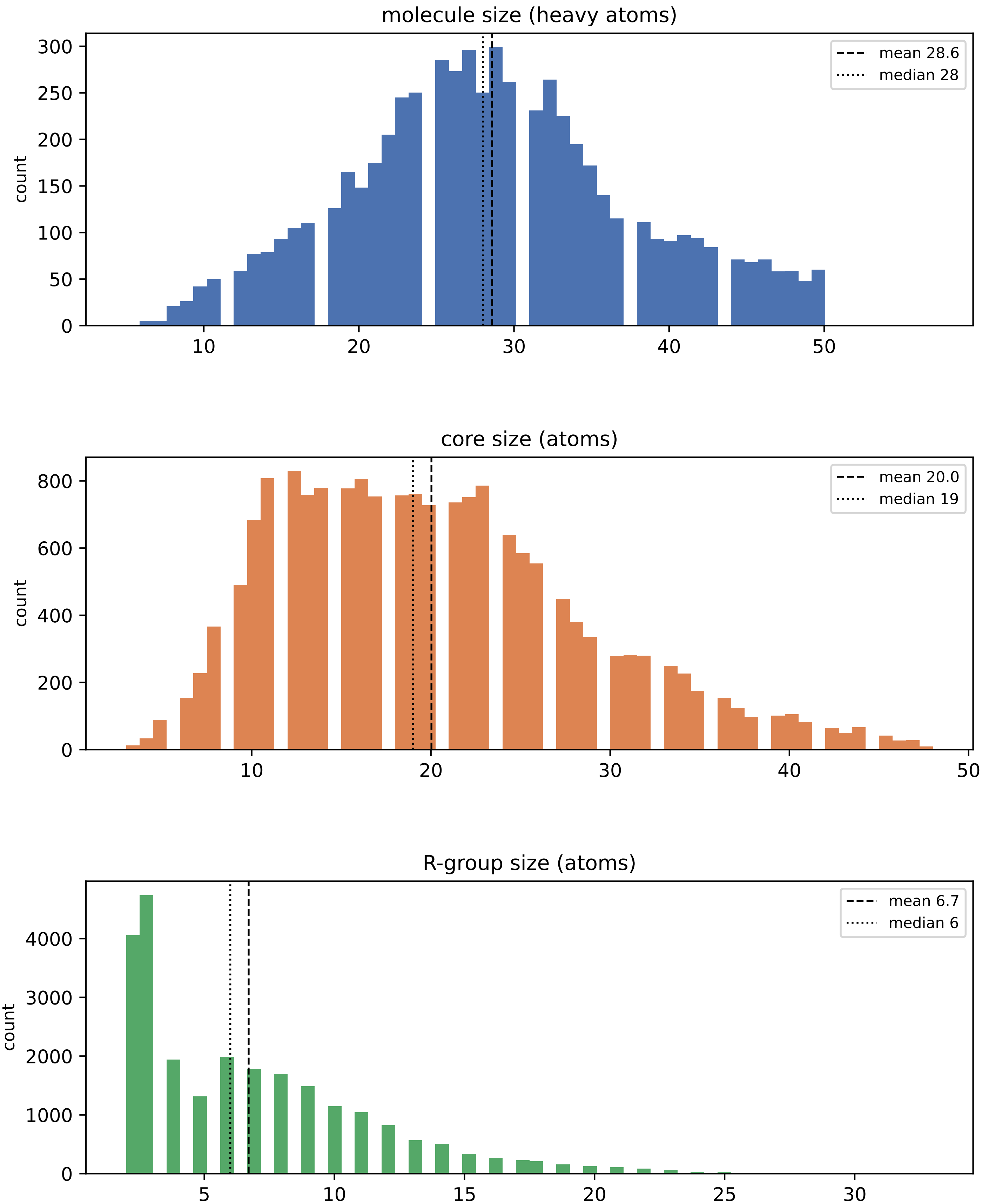
k>=2 is what the islinked subset space and the core-decoration objective have to work with.

C · Augmentation space

scheme	size
sampling_unit = molecule	374,454 instances/epoch
sampling_unit = decomposition	1,085,929 instances/epoch (2.9x)
full (mol, core, islinked) space	2,416,143 (6.5 per molecule)

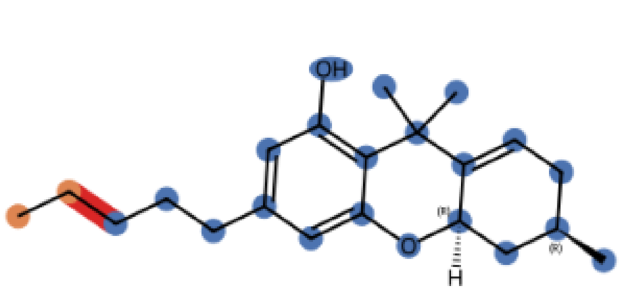
MolPLA's two stages: one molecule yields many cores; one core yields 2^k - 1 islinked subsets.

D · Size distributions

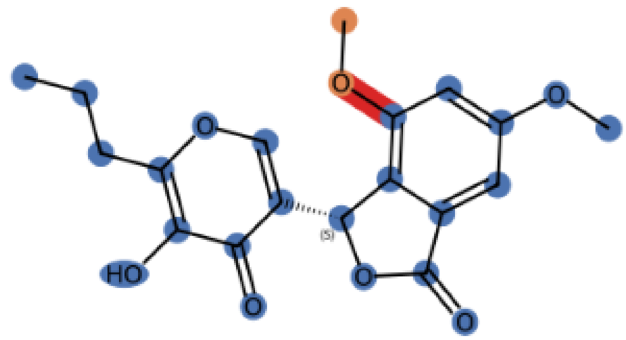


E · Example decompositions (ratio 0.3333333333333333)

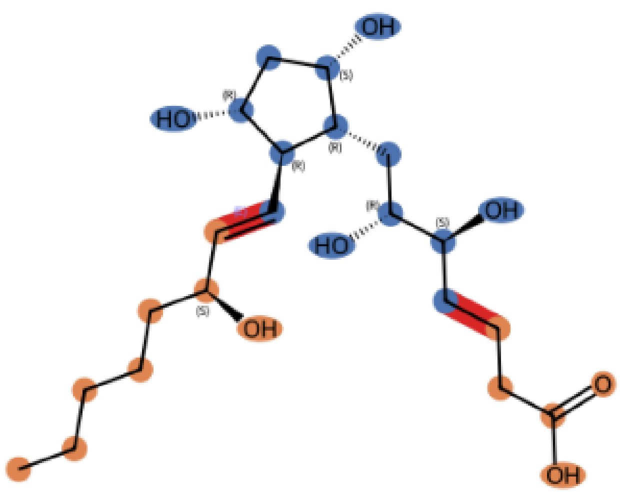
CNP0000072   k=1   core 21 atoms, R-groups [2]



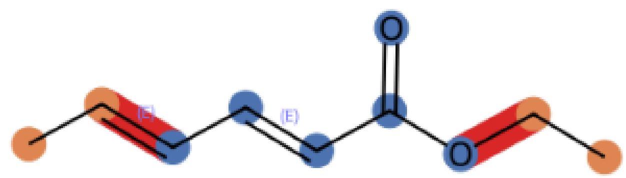
CNP0000158   k=1   core 23 atoms, R-groups [2]



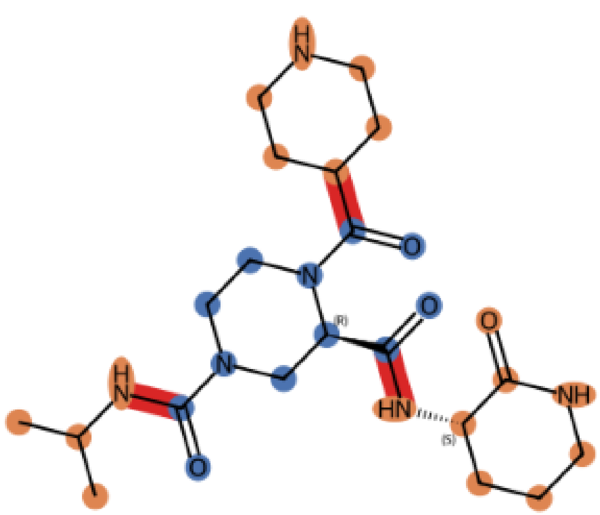
CNP0001741   k=2   core 14 atoms, R-groups [8, 5]



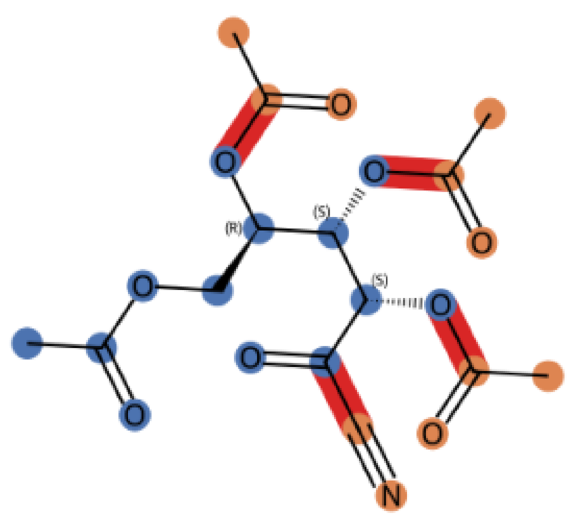
CNP0002481   k=2   core 6 atoms, R-groups [2, 2]



CNP0013440   k=3   core 12 atoms, R-groups [4, 6, 8]

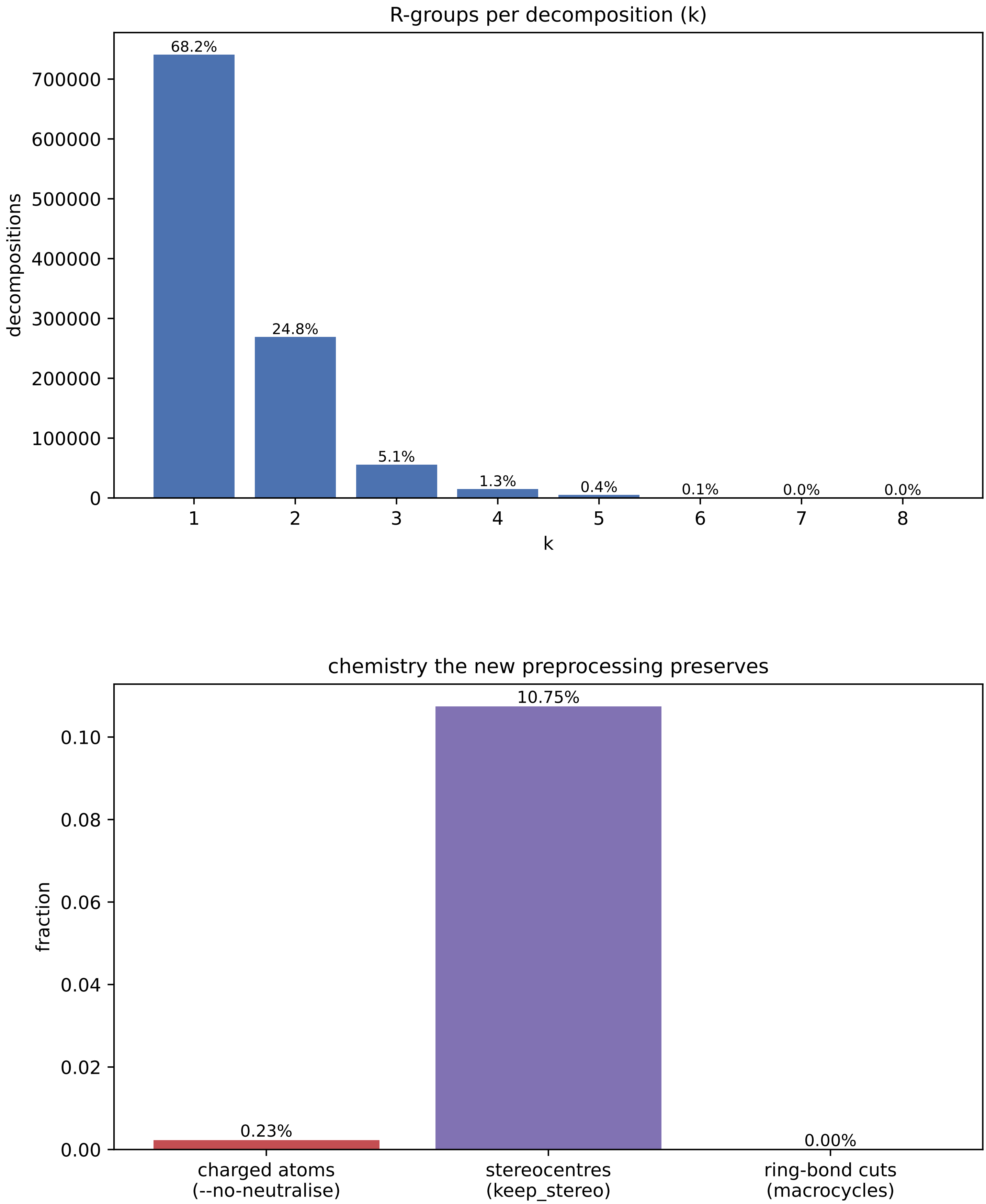


CNP0075988   k=4   core 13 atoms, R-groups [3, 3, 3, 2]

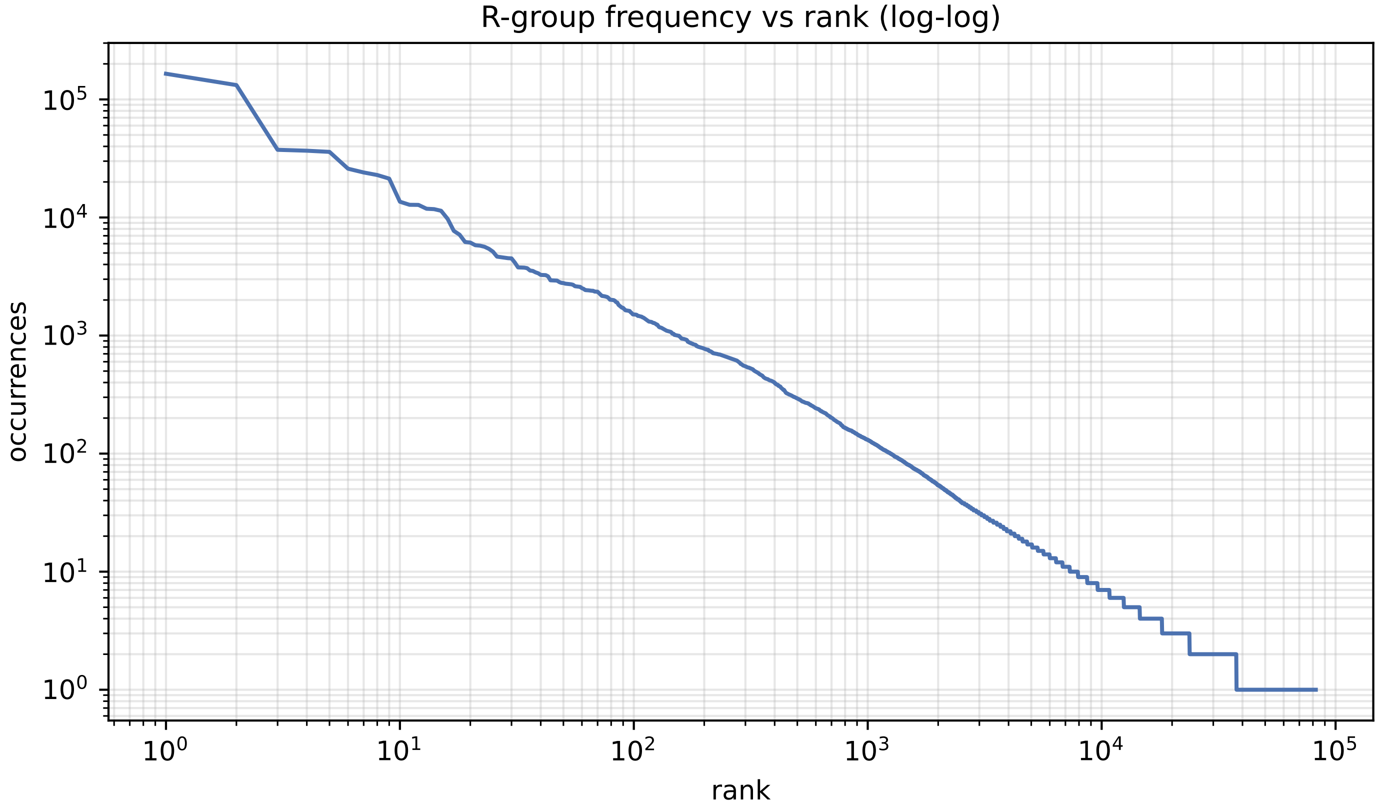


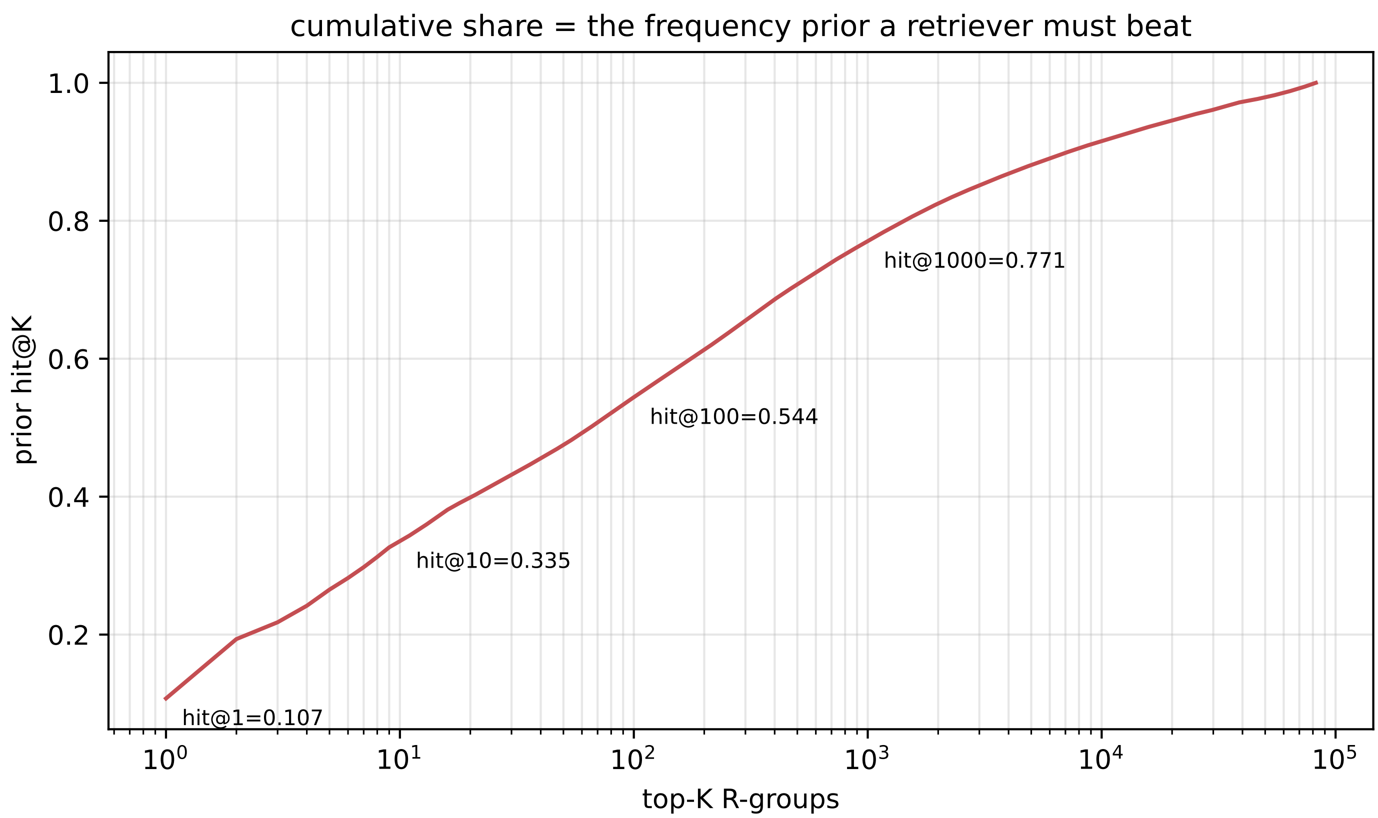
blue = core   orange = R-groups   red = cut bonds

F · Decomposition shape & preserved chemistry



G · R-group library skew





H · R-group library

quantity	value
distinct R-groups	82,312
occurrences	1,536,136
effective size (exp H)	853
top-1 share	10.74%
top-10 cover	33.5%
top-100 cover	54.4%

Effective size, not the raw count, is the number of classes retrieval actually chooses between.

I · Most frequent R-groups

count	share	SMILES
164,990	10.74%	*OC
132,252	8.61%	*C(C)=O
37,423	2.44%	*c1ccccc1
36,780	2.39%	*[CH2]C
35,945	2.34%	*C(C)C
25,893	1.69%	*[CH2]O
24,056	1.57%	*[CH](C)C
22,885	1.49%	*OC(C)=O
21,334	1.39%	*C(=O)O
13,597	0.89%	*[CH2]c1ccccc1
12,806	0.83%	*[CH2]CC
12,781	0.83%	*C(=O)c1ccccc1
11,856	0.77%	*c1ccc(OC)cc1
11,764	0.77%	*O[C@@H]1O[C@H](CO)[C@@H](O)[C@H](O)[C
11,396	0.74%	*OCC
9,724	0.63%	*c1ccc(O)cc1
7,698	0.50%	*[CH]=O
7,144	0.47%	*[CH2]C(=O)O
6,196	0.40%	*N1CCOCC1
6,129	0.40%	*[CH]=C

J · Instance-level common-R-group filter

quantity	value
percentile	99.99
count threshold	19,546
hashes flagged common	9 of 82,312
their share of occurrences	32.7%
single-R-group instances rejected	8,239/24,795 (33.2%)
mean R-group size, rejected	2.84 atoms
mean R-group size, kept	8.63 atoms

Filters INSTANCES, not the library: the retrieval target space and its frequency prior are unchanged. At k=1 the rule reduces to 'drop if the R-group is common', so rejection is far heavier than MolPLA saw.